Week 5 Feasibility Memo: AI-Assisted Triage Data Exploration

Vishal Baboolal | CariSurg MedTech Pathways Programme | Dataset: yaleemmlc_admissionprediction_triage.csv

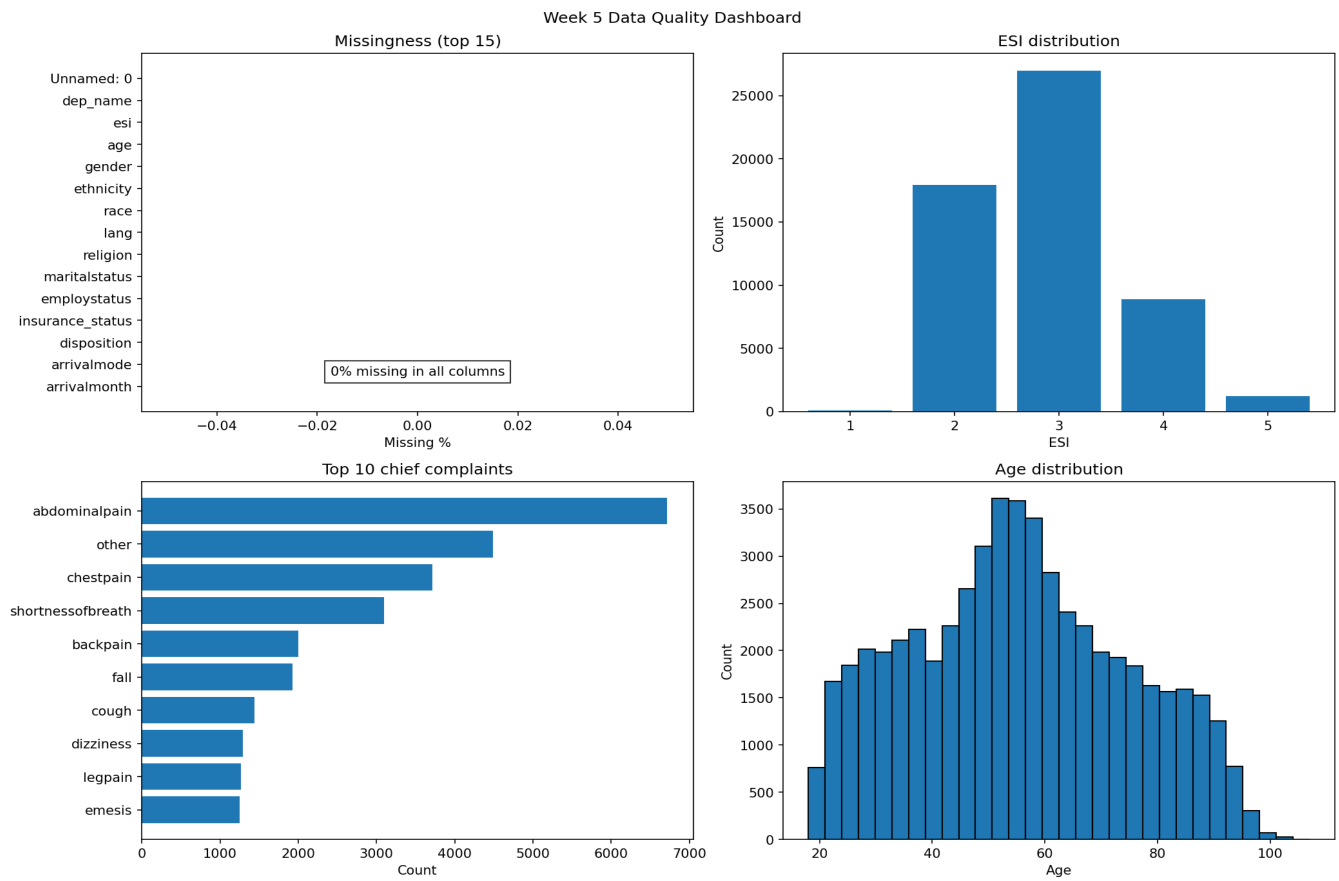
One-sentence verdict

Proceed with caveats: this dataset is suitable for a baseline emergency department triage model because it has 55,121 adult ED records, complete ESI labels, vital signs, and 200 chief-complaint flags. It should not be used clinically until leakage variables are removed, extremes are audited, fairness is checked, and local validation is completed.

Dataset summary

Item	Finding
Records / features	55,121 patient records; 226 raw columns; 225 columns after removing the original index.
Target	ESI level: 1 = most urgent, 5 = least urgent. ESI 3 is 49.0% of records; ESI 1 is 0.14%.
Age / sex	18-107 years; median 55; 57.6% female.
Race / ethnicity	53.4% White/Caucasian; 29.0% Black/African American; 17.9% Hispanic/Latino.
Arrival / complaints	41.4% car; 33.7% ambulance; 200 complaint flags; abdominal pain is most common (12.2%).
Data quality checks	0 missing values; 0 duplicate rows after ignoring original index; vital-sign extremes present.

Figure 1. Data-quality dashboard: missingness, ESI balance, top complaints and age profile



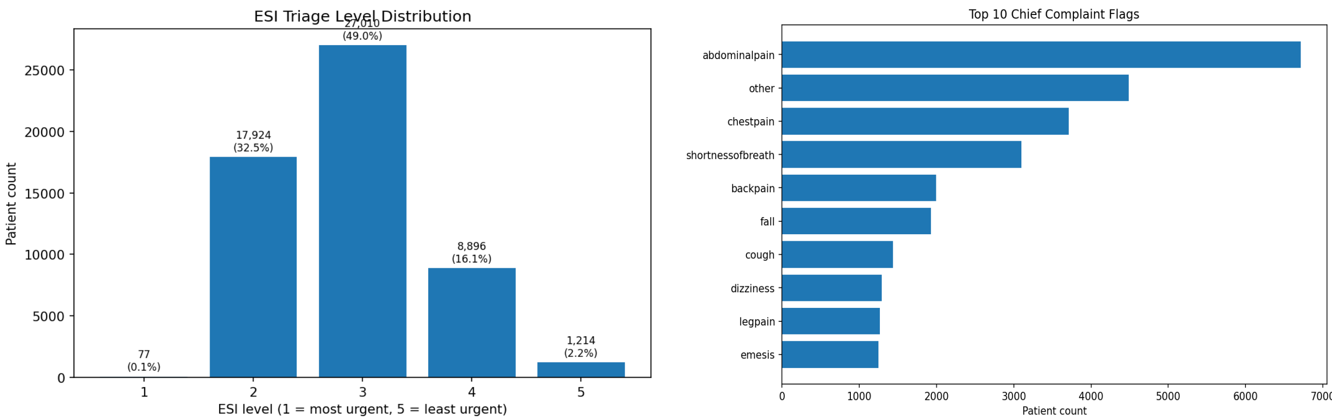
Top 3 quality concerns

Concern	Clinical meaning	Action before modelling
Leakage risk	Disposition and previous disposition are post-triage variables. If used, the model would learn information that is not available at the triage desk.	Exclude all post-triage variables and audit timestamps before training.
Target imbalance	ESI 1 is only 0.14% while ESI 3 is 49.0%. Accuracy alone could hide poor high-acuity detection.	Report class-specific recall, macro-F1, confusion matrix and ESI 1/2 recall.
Extreme values / transferability	Vital signs include extreme but possibly real values (e.g., HR 221, SBP 266, RR 66, Temp 106 F, glucose 1066). US-style insurance/arrival fields may not transfer cleanly to a Caribbean hospital.	Flag implausible values for review and validate at the intended deployment site.

Top 3 reasons to proceed

Reason	Why it matters
Large sample with complete target	55,121 records with complete ESI labels support robust exploratory analysis and a baseline model.
Clinically meaningful variables	Age, arrival mode, vital signs, oxygen-device status, glucose and chief complaints match what clinicians use at triage.
Rich chief-complaint flags	The 200 complaint flags provide clinical context for presentations such as chest pain, shortness of breath, altered mental status and suicidal ideation.

Figure 2. Target distribution and top chief complaints



Top-10 feature shortlist

Rank	Feature	Assoc.	Clinical reasoning
1	triage_vital_o2_device	-0.219	Patients needing supplemental oxygen are more likely to require urgent review; it has one of the strongest associations with lower ESI acuity level.
2	triage_vital_o2	0.159	Low oxygen saturation directly signals hypoxia and respiratory compromise, which should influence triage urgency.
3	age	-0.238	Older adults have higher risk of deterioration and admission; age shows the strongest demographic association with ESI.
4	cc_chestpain	-0.164	Chest pain may represent acute coronary syndrome and is one of the most common high-acuity complaints.
5	cc_shortnessofbreath	-0.15	Shortness of breath can indicate respiratory failure, heart failure, asthma/COPD exacerbation or sepsis.
6	triage_vital_hr	-0.073	Tachycardia or bradycardia can indicate shock, arrhythmia, pain, fever or physiological stress.
7	triage_glucose	-0.111	Abnormal glucose can indicate diabetic emergencies or severe metabolic illness.
8	cc_alteredmentalstatus	-0.132	Altered mental status is clinically high risk and associated with urgent neurologic, toxicologic, septic or metabolic causes.
9	triage_vital_rr	-0.056	Respiratory rate is an early marker of deterioration and respiratory/metabolic stress.
10	cc_suicidal	-0.143	Suicidal ideation is a safety-critical presentation requiring urgent mental health risk assessment.

Note: Lower ESI means higher urgency; negative associations indicate relationship with more urgent triage levels.

Caveats and final recommendation

This dataset is good enough for baseline modelling and feasibility testing, but not for direct clinical deployment. Week 6 should train only on variables available at or before triage, remove leakage columns, preserve clinically plausible extremes, and evaluate performance by ESI class and subgroup.

Recommended next step: build an offline baseline model and report confusion matrix, macro-F1, ESI 1/2 recall, calibration and subgroup performance across age, race, ethnicity, sex, language, insurance and arrival mode. Any future use should begin as a silent pilot before displaying AI output to triage staff.

Supporting files supplied with this memo: commented notebook, cleaned dataset, missingness table, outlier table, top complaints table, top-10 feature table, and visualization dashboard.